Introduction

ETS models — Error, Trend, Seasonal — express the exponential-smoothing family in a unified state-space form. Each component (error, trend, seasonal) is independently classified as additive, multiplicative, or absent, giving a small library of well-understood models. The state-space framework allows proper likelihood-based estimation, AIC-based model selection, and principled prediction intervals derived from the underlying stochastic structure rather than ad-hoc heuristics.

Prerequisites

A working understanding of exponential smoothing (simple, Holt's, Holt-Winters), state-space models, and the additive vs multiplicative distinction for time-series components.

Theory

The notation ETS(E, T, S) classifies each component:

- **Error**: A (additive) or M (multiplicative).
- **Trend**: N (none), A (additive), Ad (damped additive), M (multiplicative), Md (damped multiplicative).
- **Seasonal**: N (none), A (additive), or M (multiplicative).

Examples: ETS(A, N, N) is simple exponential smoothing; ETS(A, A, N) is Holt's linear trend; ETS(A, A, A) is additive Holt-Winters; ETS(M, A, M) gives multiplicative errors plus additive trend plus multiplicative seasonality. `forecast::ets()` enumerates valid candidates and selects by AIC.

The state-space form provides exact likelihood, AIC for model selection, and analytic (or simulation-based) prediction intervals. Multiplicative components require strictly positive data; the AIC-based selection avoids choosing an invalid model.

Assumptions

The chosen ETS specification matches the data's components; for prediction intervals, the innovations are approximately Normal (or simulated from the fitted residual distribution).

R Implementation

```r
library(forecast)

x <- AirPassengers

fit <- ets(x)
summary(fit)
autoplot(fit)

fc <- forecast(fit, h = 24)
autoplot(fc)

# Residual diagnostics
checkresiduals(fit)
```

Output & Results

`summary(fit)` returns the chosen model code (e.g., "M, A, M"), smoothing parameters α, β, γ, AIC, and BIC. `forecast()` produces point forecasts and 80 % / 95 % prediction intervals from the fitted state-space form. `checkresiduals()` provides Ljung-Box and graphical diagnostics.

Interpretation

A reporting sentence: "`ets()` selected ETS(M, A, M) for the airline series ($\alpha = 0.34$, $\beta = 0.01$, $\gamma = 1.0$, AIC = 1{,}393); the model captures the upward trend and the multiplicative seasonality whose amplitude grows with the level. Standardised residuals passed Ljung-Box at lag 24 ($p = 0.61$)." Always state the chosen ETS specification and the smoothing parameters.

Practical Tips

- `ets()` automatically selects the optimal ETS variant by AIC; override with `model = "AAA"` etc. when a specific form is required for replication.
- Multiplicative components require strictly positive data; AIC-based selection avoids invalid choices automatically.
- For complex multi-seasonality (e.g. weekly + yearly), TBATS (`forecast::tbats()`) extends ETS with multiple seasonal layers and Box-Cox transformation.
- ETS prediction intervals can be very wide at long horizons because the multiplicative variance compounds; for a more nuanced view, simulate many sample paths via `forecast(fit, simulate = TRUE, npaths = 1000)`.
- Compare ETS against ARIMA via rolling-origin cross-validation; both are strong baseline forecasters and the better choice depends on the series.
- For damped trend (which extrapolates more conservatively at long horizons), use `model = "AAdA"` or let `ets()` discover it through AIC.

R Packages Used

`forecast::ets()` for the canonical ETS implementation with AIC-based selection; `fable::ETS()` for the modern tidyverts equivalent; `forecast::tbats()` for multiple seasonalities and Box-Cox transformation; `smooth::es()` for advanced ETS variants including ETS plus regressors.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Time series basics